LMECA2323 – Turbomachinery

Project – 2026

Characterization of a high-pressure turbine stage
for an industrial power production gas turbine

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Chapter 1

Introduction

A turbomachine produces or absorbs energy through a change of enthalpy in a continuously flowing stream, transferring work through a rotating shaft. In a gas turbine the turbine extracts work from the hot combustion products by expansion: the flow accelerates at the expense of pressure, and a large part of the extracted power is used to drive the compressor through the shaft, the remainder being the useful (net) output.

Designing a turbine stage couples thermodynamics, kinematics (velocity triangles), annulus geometry and mechanical constraints [1]. The procedure followed in this work is the classical three-step chain of preliminary design:

- **1-D mean-line design** – thermodynamic and velocity-triangle analysis on the mean streamline;
- **Rotor-blade design** – conversion of the flow angles into a blade profile (chord, stagger, pitch, blade count);
- **CFD evaluation** – meshing and RANS solution of the resulting cascade to verify the loading and exit flow.

These steps are not executed once and in isolation: the design coefficients (stage loading ψ, flow coefficient ϕ and degree of reaction r) are iterated until the resulting stage satisfies a set of aerodynamic and mechanical design guidelines. This report focuses on the **first stage** of a four-stage high-pressure turbine sized for the conditions of an F-class industrial gas turbine (representative of the Siemens SGT5-4000F), delivering 330 MW of shaft power at a mass flow of 750 kg/s.

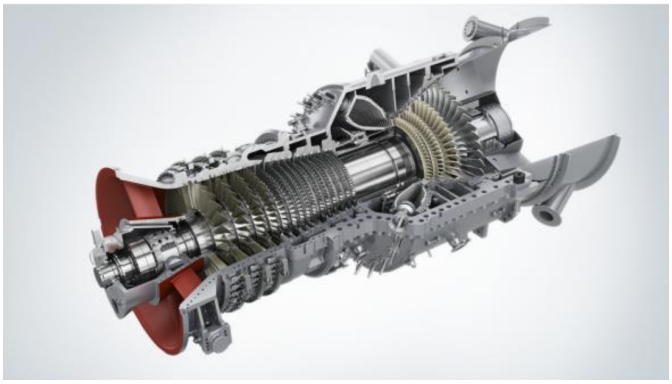


Figure 1.1: Industrial gas turbine of the SGT5-4000F class (cutaway).

The whole design chain has been implemented in Python (the accompanying notebook): the mean-line solver, the ParaBlade-based blade construction [6] and the GMSH [8] + SU2 [7] cascade CFD. All numerical results, tables and figures in this report are produced directly by that code.

Chapter 2

1-D Mean-Line Design

The 1-D mean-line design is the simplified modelling strategy used in the early stages of turbine design. The flow is assumed axisymmetric and is averaged along the radial direction, so that the whole stage is represented by the conditions on the mean streamline. With this simplification the stage power, efficiency and blade loading can be evaluated quickly and the design coefficients iterated.

The stage is divided into three stations: station 1 (stator inlet), station 2 (stator exit / rotor inlet) and station 3 (rotor exit).

2.1 Input Data

The following quantities are imposed by the project statement:

- Working fluid: calorically perfect gas, $\gamma = 4/3 \approx 1.33$, $R = 287\,\mathrm{J/kg/K}$.
- Turbine mass flow $\dot{m} = 750\,\mathrm{kg/s}$; fuel-to-air ratio $f_{ar} = 0.025$.
- Stage (extracted) shaft power $P = 330\,\mathrm{MW}$ (the turbine also drives the compressor).
- Overall compressor pressure ratio $\Pi_C = 20$.
- Compressor admission $T_{adm} = 15\,^\circ\mathrm{C}$, $P_{adm} = 101\,325\,\mathrm{Pa}$, isentropic efficiency $\eta_{is,C} = 0.92$.
- Combustion-chamber Mach number $M_{comb} = 0.15$.
- Turbine inlet static temperature (F-class) $TIT = 1400\,^\circ\mathrm{C}$.
- Inlet yaw angle $\alpha_1 = 0^\circ$ (fully axial); rotational speed $N_{rot} = 3000\,\mathrm{rpm}$ (synchronous generator).
- Blade peripheral speed limited to $U_m = \omega R_m \leq 350\,\mathrm{m/s}$; mechanical constraint $AN^2 < 3 \times 10^7\,\mathrm{m^2\,rpm^2}$.
- Constant mean radius R_{mean} across the stage; fixed blading efficiencies $\eta_{S,R} = 0.90$.

2.2 Design Procedure

2.2.1 Inlet Condition Calculations

The specific heats are obtained from γ and R:

$$c_p = \frac{\gamma R}{\gamma - 1} = 1156.70\,\mathrm{J/kg/K}, \qquad c_v = \frac{R}{\gamma - 1} = 869.70\,\mathrm{J/kg/K}. \tag{2.1}$$

Only air flows through the compressor, so its mass flow differs from the turbine:

$$\dot{m}_c = \frac{\dot{m}_T}{1.025} = 731.71\,\mathrm{kg/s}. \tag{2.2}$$

With $\gamma_C = 1.4$ for the compressor, the absorbed compressor power is

$$P_c = \frac{1}{\eta_{is}\,\eta_m}\,\dot{m}_c\,c_{p,C}\,T_{adm}\left[\Pi_C^{\frac{\gamma_C - 1}{\gamma_C}} - 1\right] = 327.82\,\mathrm{MW}, \tag{2.3}$$

taking a mechanical efficiency $\eta_m = 0.95$. The compressor exit (turbine inlet) static pressure follows from the pressure ratio,

$$P_1 = \Pi_C\,P_{adm} = 2\,026\,500\,\mathrm{Pa}. \tag{2.4}$$

The turbine inlet stagnation properties are obtained from the combustion-chamber Mach number:

$$P_{01} = P_1\left(1 + \tfrac{\gamma-1}{2}M_{comb}^2\right)^{\frac{\gamma}{\gamma-1}} = 2\,056\,992\,\mathrm{Pa}, \tag{2.5}$$

$$T_{01} = T_1\left(1 + \tfrac{\gamma-1}{2}M_{comb}^2\right) = 1679.21\,\mathrm{K}, \tag{2.6}$$

with $T_1 = TIT = 1673\,\mathrm{K}$.

2.2.2 Required P_3 Calculations

The total power to be produced by the four-stage turbine is the sum of the net shaft power and the compressor power, so the power per stage is

$$P_{stage} = \frac{P + P_c}{4} = 164.46\,\mathrm{MW}. \tag{2.7}$$

The corresponding real specific work and isentropic enthalpy drop (with $\eta_{TT} = 0.87$) are

$$\Delta H_{real} = \frac{P_{stage}}{\dot{m}} = 219.27\,\mathrm{kJ/kg}, \qquad \Delta H_{is} = \frac{\Delta H_{real}}{\eta_{TT}} = 243.64\,\mathrm{kJ/kg}. \tag{2.8}$$

The isentropic exit stagnation state and stage pressure ratio are

$$T_{03,is} = T_{01} - \frac{\Delta H_{is}}{c_p} = 1468.58\,\mathrm{K}, \quad P_{03,is} = 1\,198\,493\,\mathrm{Pa}, \quad \Pi_{TT,is} = \frac{P_{01}}{P_{03,is}} \approx 1.72. \tag{2.9}$$

Assuming an exit Mach number $M_3 = 0.3$ (within the recommended 0.3–0.4 band) closes the static exit state and gives a first estimate of P_3, which is later refined by the iteration of Section 2.5.1.

2.2.3 Assumptions of Design Parameters

Three coefficients are chosen and iterated to meet the design guidelines:

Stage loading $\psi = \Delta H_{real}/U^2$. For a high-pressure stage ψ is typically 1.5–2. It controls the width of the velocity triangles, the turning and the exit Mach number. Fixing $\psi = 1.8$ gives a blade speed

$$U_m = \sqrt{\frac{\Delta H_{real}}{\psi}} = 349.03\,\mathrm{m/s}, \tag{2.10}$$

just within the 350 m/s mechanical limit.

Flow coefficient $\phi = V_x/U$. The flow coefficient sets the height of the velocity triangles. Using a Smith chart [3] at the chosen efficiency and ψ, a value $\phi = 0.5$ is selected, giving an axial velocity $V_x = 174.5\,\mathrm{m/s}$.

Degree of reaction $r = (h_2 - h_3)/(h_1 - h_3)$. For a cooled high-pressure stage the reaction is kept low, 0.35–0.5; here $r = 0.35$ is adopted.

2.3 Thermodynamic Quantities

With the inlet state and the three coefficients fixed, the thermodynamic and kinematic quantities are computed at every station. Station 2 follows from the stator expansion ($V_{x2} = U\phi$, $P_2 = r(P_1 - P_3) + P_3$, stator efficiency η_S); the relative frame (W_2, β_2, T_{02R}, P_{02R}) is then formed, and the rotor exit (station 3) is obtained from the relative isentropic expansion to P_3 with rotor efficiency η_R. Table 2.1 collects the converged values.

Table 2.1: Thermodynamic and kinematic quantities at the three stations.

Quantity	Station 1	Station 2	Station 3
Static pressure P [Pa]	2026500	1428997	1107265
Total pressure (abs.) P_0 [Pa]	2056992	1954202	1166813
Rel. total pressure P_{0R} [Pa]	–	1521338	1472070
Static temperature T [K]	1673.0	1553.7	1470.4
Total temperature (abs.) T_0 [K]	1679.2	1679.2	1489.6
Rel. total temperature T_{0R} [K]	–	1578.1	1578.1
Absolute velocity V [m/s]	174.5	538.8	211.0
Axial velocity V_x [m/s]	174.5	174.5	174.5
Tangential velocity V_u [m/s]	0.0	509.7	118.5
Relative velocity W [m/s]	–	237.2	499.1
Yaw angle α [°]	0.00	71.10	34.18
Relative angle β [°]	–	42.64	69.53
Absolute Mach M [-]	0.22	0.70	0.30
Relative Mach M_w [-]	–	0.31	0.67

The stator-exit Mach $M_2 = 0.70$ and the relative exit Mach $M_{w3} = 0.67$ are consistent with a low-reaction high-pressure stage in which the bulk of the acceleration occurs in the stator.

2.4 Velocity Triangles

The velocity triangles at stations 2 and 3 are built from the blade speed U, the absolute velocity V and the relative velocity W, together with the flow angle α and the blade angle β. They are plotted in Fig. 2.1 and summarised in Table 2.2.

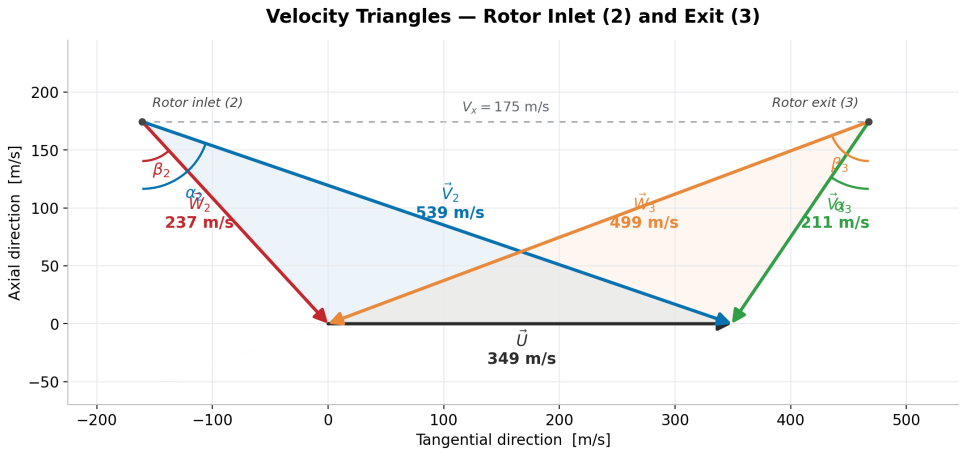


Figure 2.1: Velocity triangles at rotor inlet (station 2) and exit (station 3).

Table 2.2: Velocity-triangle parameters.

Parameter	Station 2	Station 3
U – blade speed [m/s]	349.03	349.03
V – absolute velocity [m/s]	538.78	210.95
W – relative velocity [m/s]	237.24	499.05
α – flow angle [°]	71.10	34.18
β – blade angle [°]	42.64	69.53

The total turning of the rotor is $\Delta\beta = \beta_2 + \beta_3 = 112.2°$, typical of a heavily loaded high-pressure rotor.

2.5 h–s Diagram

The enthalpy–entropy diagram (Fig. 2.2) summarises the expansion: the isentropic, static and stagnation enthalpies at each station and the kinetic energy terms $V^2/2$ and $W^2/2$. The enthalpy ladder is listed in Table 2.3.

Table 2.3: Enthalpy ladder of the stage (J/kg).

Quantity	Value	Quantity	Value
$H_{01} = H_{02}$	1942338	h_3	1700813
h_1	1935154	h_{3s}	1686977
h_2	1797198	h_{3ss}	1665655
h_{2s}	1781071	H_{03}	1723063
H_{0R}	1825338	H_{03ss}	1687906

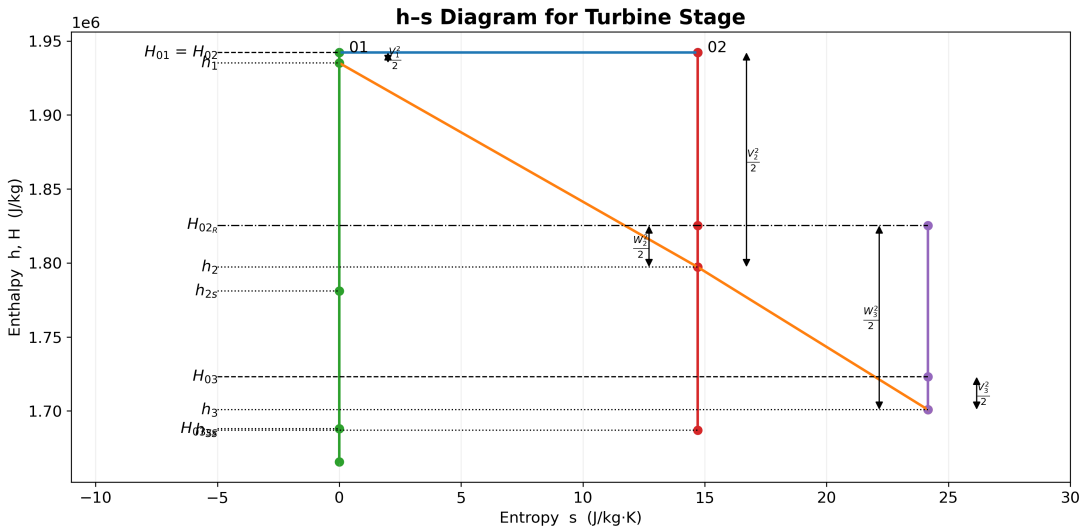


Figure 2.2: h–s diagram of the first turbine stage.

2.5.1 Comparison of Enthalpy (closure of the design)

The exit static pressure P_3 is not known a priori. The stage enthalpy drop computed kinematically from the velocity triangles, $\Delta H_0 = c_p(T_{01} - T_{03})$, must equal the value $\Delta H_{real} = P_{stage}/\dot{m}$ imposed by the power balance. A Newton–Raphson loop adjusts P_3 until the two agree:

$$\Delta H_0 < \Delta H_{real} \Rightarrow P_3' < P_3, \quad \Delta H_0 > \Delta H_{real} \Rightarrow P_3' > P_3, \quad \Delta H_0 = \Delta H_{real} \Rightarrow \text{OK}.$$

The loop converges in 33 iterations to $P_3 = 1\,107\,265\,\text{Pa}$, with $\Delta H_0 = \Delta H_{real} = 219.27\,\text{kJ/kg}$ (residual $< 1 \times 10^{-5}\,\text{kJ/kg}$).

2.6 Annulus Sizing

Annulus sizing fixes the radial extent of the blade row. With a shrouded end-wall assumption (equal hub and tip contour angles ε), the areas follow from mass conservation, $A = \dot{m}/(\rho V_x)$, and the heights from $h = A/(\pi D_m)$. The mean radius is set by the blade speed, $R_m = 60\,U_m/(2\pi N) = 1.11\,\text{m}$. Table 2.4 lists the results.

Table 2.4: Annulus geometry (constant mean radius).

Quantity	**Value**
Mean radius R_m [m]	1.110
Span h_2 [mm]	192.1
Span h_3 [mm]	234.6
Tip radius (st. 3) R_{T3} [m]	1.228
Hub radius (st. 3) R_{H3} [m]	0.994
End-wall flare ε [°]	15.90
AN^2 [m^2 rpm^2]	1.341×10^7

The end-wall flare angle, obtained from the span growth over the axial chord, $\varepsilon = \tan^{-1}\big((h_3 - h_2)/c_x\big) = 15.90°$, stays below the 15–20° range that bounds end-wall separation.

2.7 Design Constraints

Table 2.5 compares the obtained values against the aerodynamic and mechanical design guidelines. Eight of the ten constraints are satisfied; two are marginally violated – the stator-exit Mach number $M_2 = 0.700$ sits exactly at the lower edge of the 0.7–0.85 band, and the span ratio $h_3/h_2 = 1.221$ slightly exceeds the recommended 1.2. Both are borderline and could be cleared by a small change of ψ, ϕ or r.

Table 2.5: Comparison of design control parameters and obtained results.

Constraint	**Design range**	**Obtained**	**Status**
M_2	0.7–0.85	0.700	FAIL (low)
β_2 [°]	< 47.5	42.64	PASS
M_{w2}	< 0.5	0.308	PASS
M_{w3}	0.65–0.8	0.666	PASS
β_3 [°]	< 75	69.53	PASS
$\Delta\beta$ [°]	110–120	112.17	PASS
M_3	0.3	0.300	PASS
α_3 [°]	< 35	34.18	PASS
h_3/h_2	< 1.2	1.221	FAIL (high)
AN^2 [m^{2}rpm^2]	$< 3 \times 10^7$	1.34×10^7	PASS

Chapter 3

Rotor Blade Design

The rotor-blade design converts the mean-line flow angles into an actual blade profile. A few geometric assumptions (aspect ratio, stagger) fix the chord and the number of blades, after which the camber line and the thickness distribution are built with ParaBlade from the inlet/exit metal angles.

3.1 Input Data Assumptions

Aspect ratio $AR = b/\bar{c}$. The aspect ratio relates the blade span b to the chord $\bar{c}$. For high-pressure rotors AR lies between 1 and 1.5; a value $AR = 1$ is adopted, which gives the best compromise for the present loading.

Stagger angle ξ. The stagger is the angle between the chord and the axial direction. It is read from the Kacker & Okapuu correlation [5] as a function of the inlet and exit relative angles $\beta_2 = 42.64°$ and $\beta_3 = 69.53°$, giving $\xi \approx 39°$.

Zweifel loading coefficient. The blade pitch (and hence the blade count) is set from the Zweifel criterion [4] with $Z = 1.0$, which fixes the optimum pitch-to-axial-chord ratio for the prescribed turning.

3.2 Design Procedure and Calculations

From the aspect ratio and the span, the chord and axial chord are obtained; the Zweifel criterion then fixes the pitch and, with the mean circumference, the number of blades:

$$N_b = \frac{2\pi R_m}{s}, \qquad s = \text{pitch}. \tag{3.1}$$

The camber line is generated between the metal angles β_2 (inlet) and β_3 (exit) and combined with a prescribed thickness distribution to produce the suction and pressure surfaces using ParaBlade [6]. The tool reproduces the prescribed analytic curvature: a check of the curvature design variable against the analytic surface curvature matches at both ends ($\kappa_{LE} = 17.61$, $\kappa_{TE} = 213.29$), confirming a consistent profile.

3.3 Main Blade Parameters

The resulting blade parameters are collected in Table 3.1; the constructed section is shown in Fig. 3.1.

Table 3.1: Main rotor-blade parameters.

Parameter	**Value**
Inlet flow angle α_2 [°]	71.10
Inlet metal (rel.) angle β_2 [°]	42.64
Exit flow angle α_3 [°]	34.18
Exit metal (rel.) angle β_3 [°]	69.53
Aspect ratio AR	1.00
Stagger angle ξ [°]	39.0
Zweifel coefficient Z	1.00
Chord c [m]	0.192
Axial chord c_x [m]	0.149
Pitch s [m]	0.170
Pitch-to-axial-chord s/c_x	1.14
Number of blades N_b	42
Max. thickness-to-chord t_{max}/c	0.20
Leading-edge radius R_{LE} [m]	0.0085
Trailing-edge radius R_{TE} [m]	0.0007
End-wall flare angle ε [°]	15.90

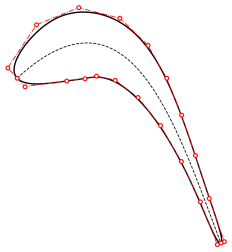


Figure 3.1: Constructed rotor-blade section (camber line and thickness distribution).

Chapter 4

Rotor Blade Construction and CFD Evaluation

The blade profile obtained in Chapter 3 is assembled into a linear cascade and evaluated with a 2-D RANS computation. This chapter describes the passage geometry, the mesh, the solver setup and convergence, and the post-processed aerodynamic results (loading, Mach/pressure fields and the exit wake).

4.1 Blade Passage Geometry

The cascade is modelled as three blades: the central (measured) blade and its two neighbours, whose facing surfaces form the upper and lower passage walls. The non-dimensional pitch and throat characterise the passage:

$$s/c_x = 1.136, \quad o/c_x = 0.805, \quad o/s = 0.709, \quad (x/c_x)_{throat} = 0.284, \tag{4.1}$$

with dimensional values $c_x = 0.149\,\mathrm{m}$, $s = 0.170\,\mathrm{m}$ and throat $o = 0.120\,\mathrm{m}$, for $N_b = 42$ blades.

4.2 Evaluation of Profile Curvature and Passage Area

The prescribed thickness distribution and the analytic curvature of the surfaces are shown in Fig. 4.3; the passage-width (area) variation along the axial direction, which contracts monotonically to the throat, is shown in Fig. 4.1. A smooth, monotonic contraction without local diffusion is desirable to avoid boundary-layer growth on the suction side.

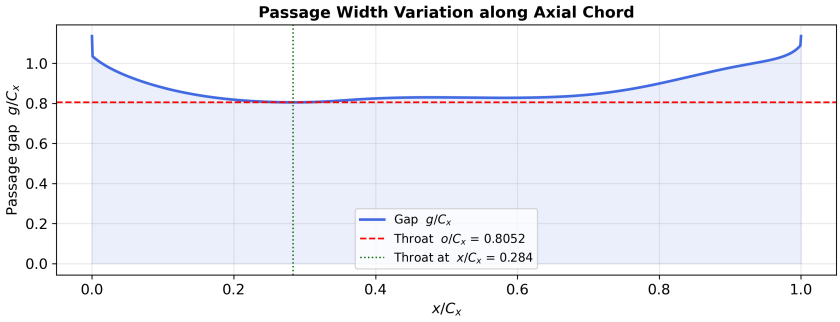
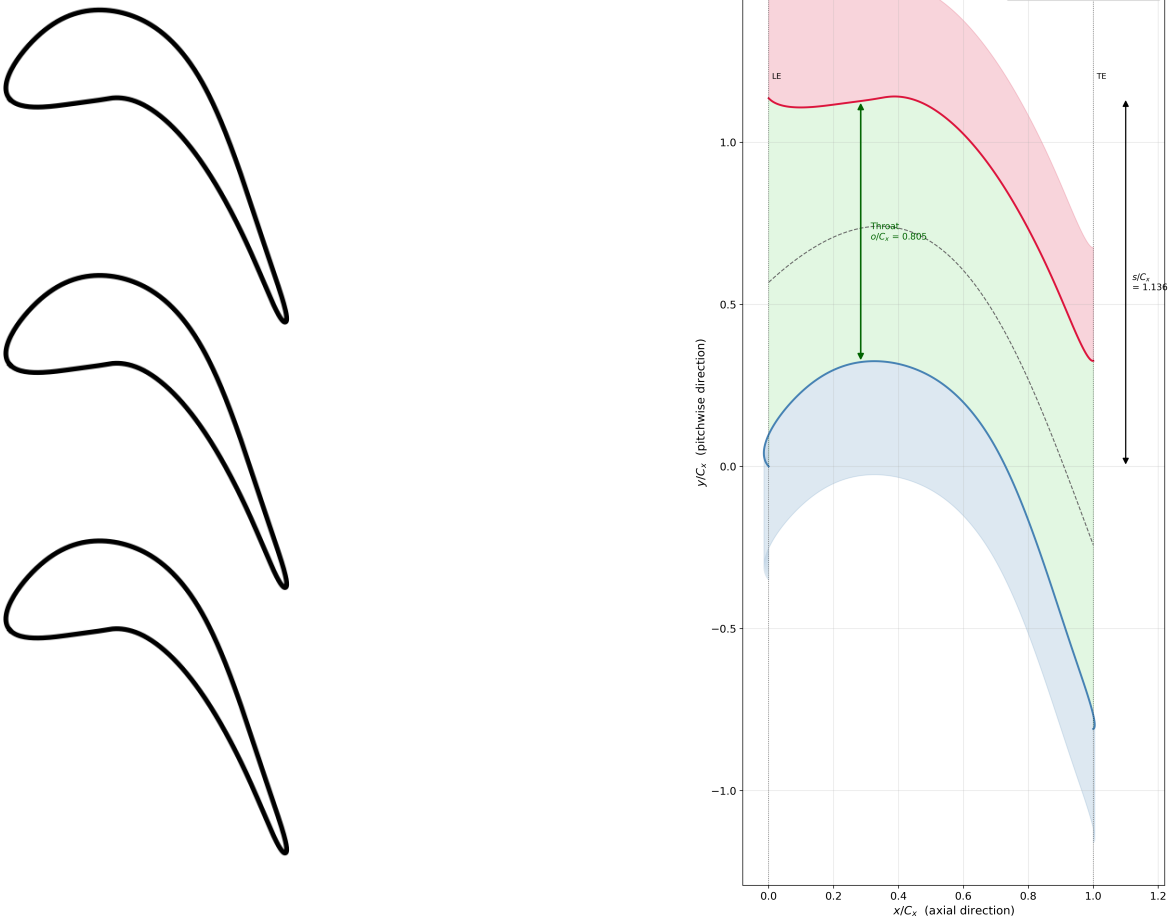


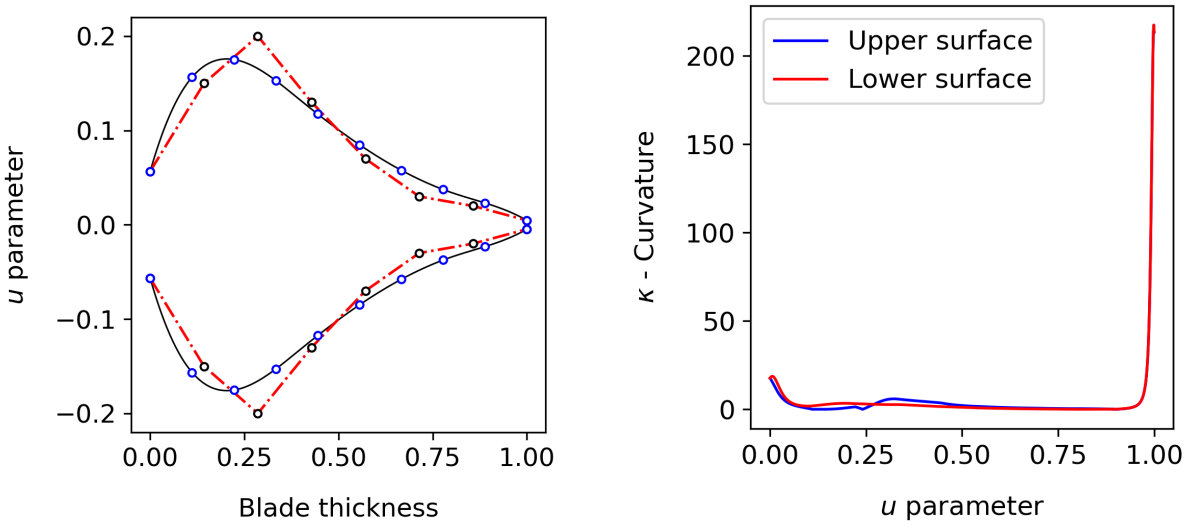
Figure 4.1: Passage-width (area) variation through the blade row.



(a) Cascade of rotor blades.

(b) Blade-to-blade passage and throat.

Figure 4.2: Rotor-blade cascade and passage geometry.



(a) Thickness distribution.

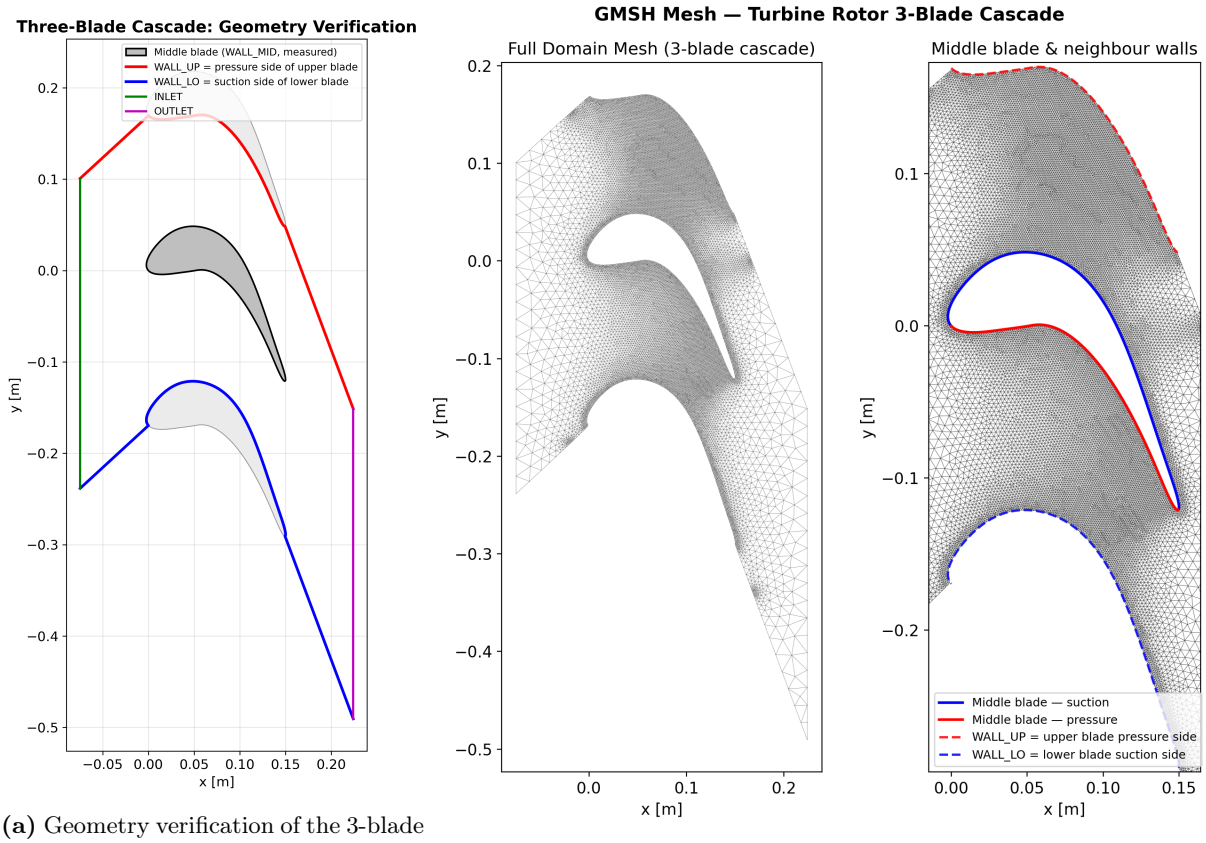
(b) Curvature κ distribution.

Figure 4.3: Profile thickness and curvature along the normalised surface coordinate.

4.3 CFD Evaluation

4.3.1 Mesh Generation

An unstructured triangular mesh is generated with GMSH [8] over the three-blade passage. The domain is bounded by the central blade (no-slip wall), the two neighbour walls (the pressure side of the upper blade and the suction side of the lower blade), and inlet/outlet planes extended half an axial chord upstream and downstream along the flow angles $+\beta_2$ and $-\beta_3$. The inlet and outlet heights are both $\approx 2s = 339.2\,\text{mm}$. A distance-based size field refines the cells near all blade walls. The mesh contains about 1.81×10^4 nodes and 3.54×10^4 elements.



(a) Geometry verification of the 3-blade domain.

(b) Generated GMSH mesh.

Figure 4.4: Three-blade cascade domain and mesh.

4.3.2 CFD Setting

The flow is solved with SU2 [7] (compressible RANS, Spalart–Allmaras turbulence model). The boundary conditions are imposed in the rotor-relative frame: stagnation inlet from station 2 (T_{02R}, P_{02R}, flow direction β_2), static back-pressure P_3 at the outlet, a no-slip adiabatic central blade and inviscid (Euler) neighbour walls. The main settings are listed in Table 4.1.

Table 4.1: SU2 CFD solver configuration.

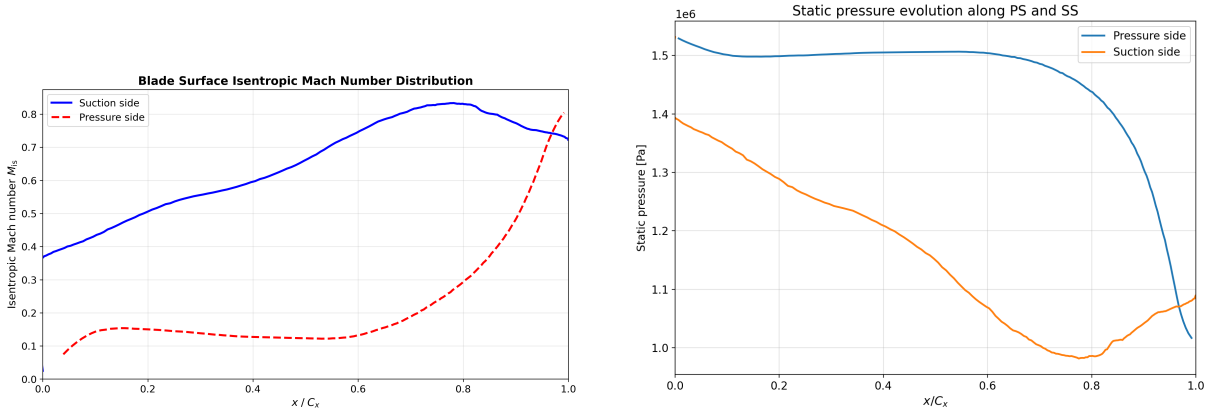
Setting	Value
Solver	RANS (compressible)
Turbulence model	Spalart–Allmaras
Inlet total temperature T_{02R} [K]	1578.1
Inlet total pressure P_{02R} [Pa]	1521338
Inlet flow angle β_2 [°]	42.64
Outlet static pressure P_3 [Pa]	1107265
Inlet relative Mach M_{w2}	0.308
Central blade (WALL_MID)	no-slip, adiabatic
Neighbour walls (WALL_UP/LO)	Euler (slip)
Convergence monitor	RMS density residual

4.3.3 CFD Convergence

The simulation is advanced until the density residual drops by several orders of magnitude (RMS[ρ] falling below -7.7), at which point the surface loading and the integral exit quantities are stationary. The solution is then used for the post-processing below.

4.4 Isentropic Static-Pressure Distribution

The blade loading is examined through the isentropic Mach number and the static pressure on the suction and pressure surfaces (Fig. 4.5). The suction-side isentropic Mach number rises to $M_{is} \approx 0.84$ near the throat before recovering, while the pressure side stays below ≈ 0.79; the static pressure spans 1.02–1.53 MPa on the pressure side and 0.98–1.52 MPa on the suction side. The loading is front-to-mid loaded with no sign of suction-side separation.



(a) Isentropic Mach number.

(b) Static pressure on PS / SS.

Figure 4.5: Blade loading: isentropic Mach and static-pressure distribution.

4.5 Colourmaps of the CFD Analysis

The Mach-number and static-pressure fields (Fig. 4.6) show the flow accelerating through the converging passage, the peak velocity on the suction side near the throat, and the smooth pressure relaxation towards the outlet, consistent with the prescribed expansion to P_3.

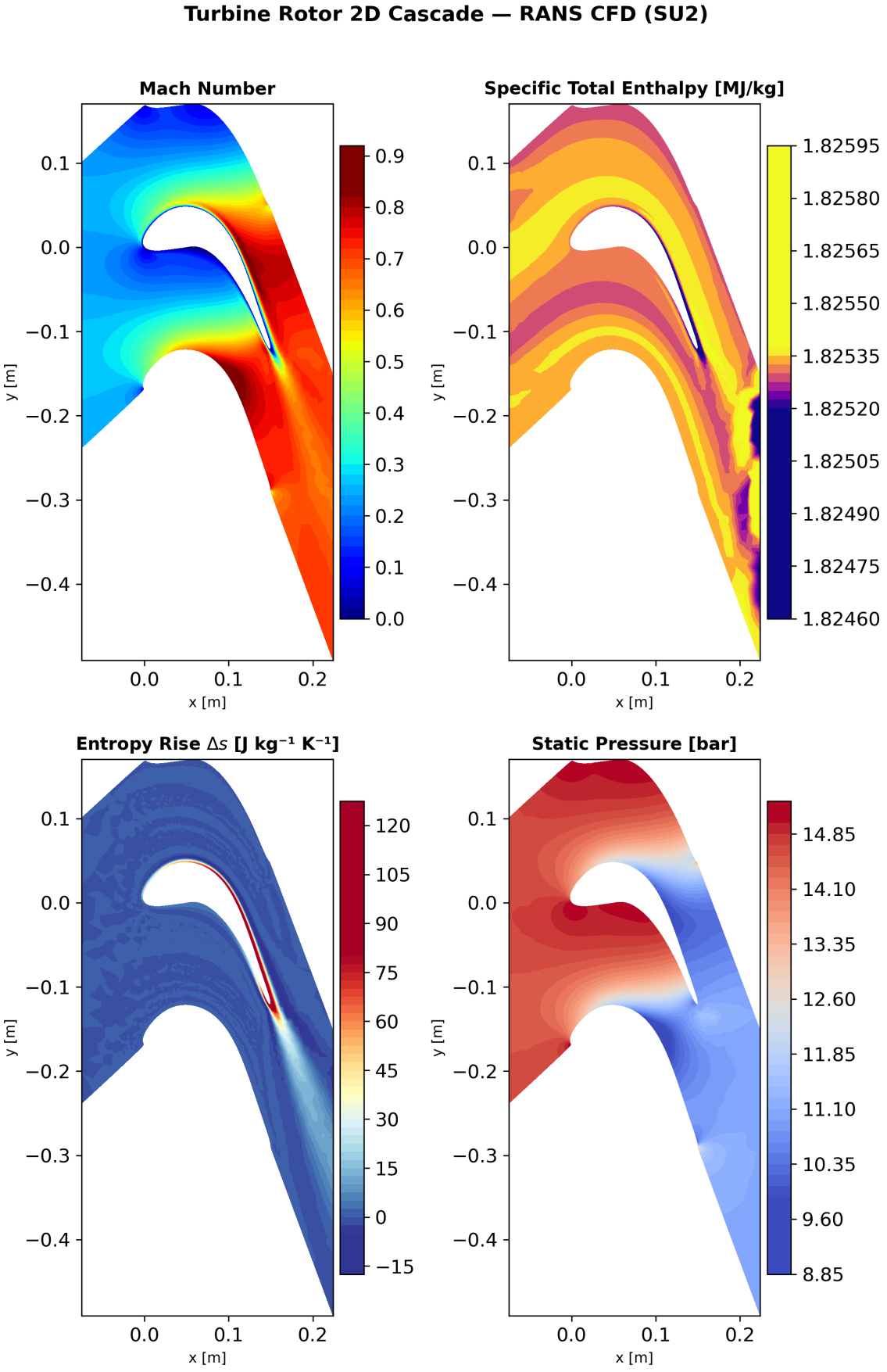


Figure 4.6: CFD colourmaps of the cascade flow: Mach number, specific total enthalpy, entropy rise and static pressure.

4.6 Velocity Profile and Wake

The pitchwise distributions one trailing-edge region downstream (Fig. 4.7) reveal the blade wake as a momentum deficit in all four quantities – total pressure, Mach number, flow angle and axial velocity. At the extraction plane $x = x_{TE} + 0.30\,c_x = 0.194\,\text{m}$ the mass-averaged total pressure is $P_t = 1.49\,\text{MPa}$ ($P_t/P_{t0} = 0.978$), the mean Mach number is 0.681 and the mean flow angle is $-70.7°$. The axial velocity drops from $\approx 191\,\text{m/s}$ in the free stream to $\approx 158\,\text{m/s}$ in the wake core (a 9.2% deficit), narrow and well-mixed.

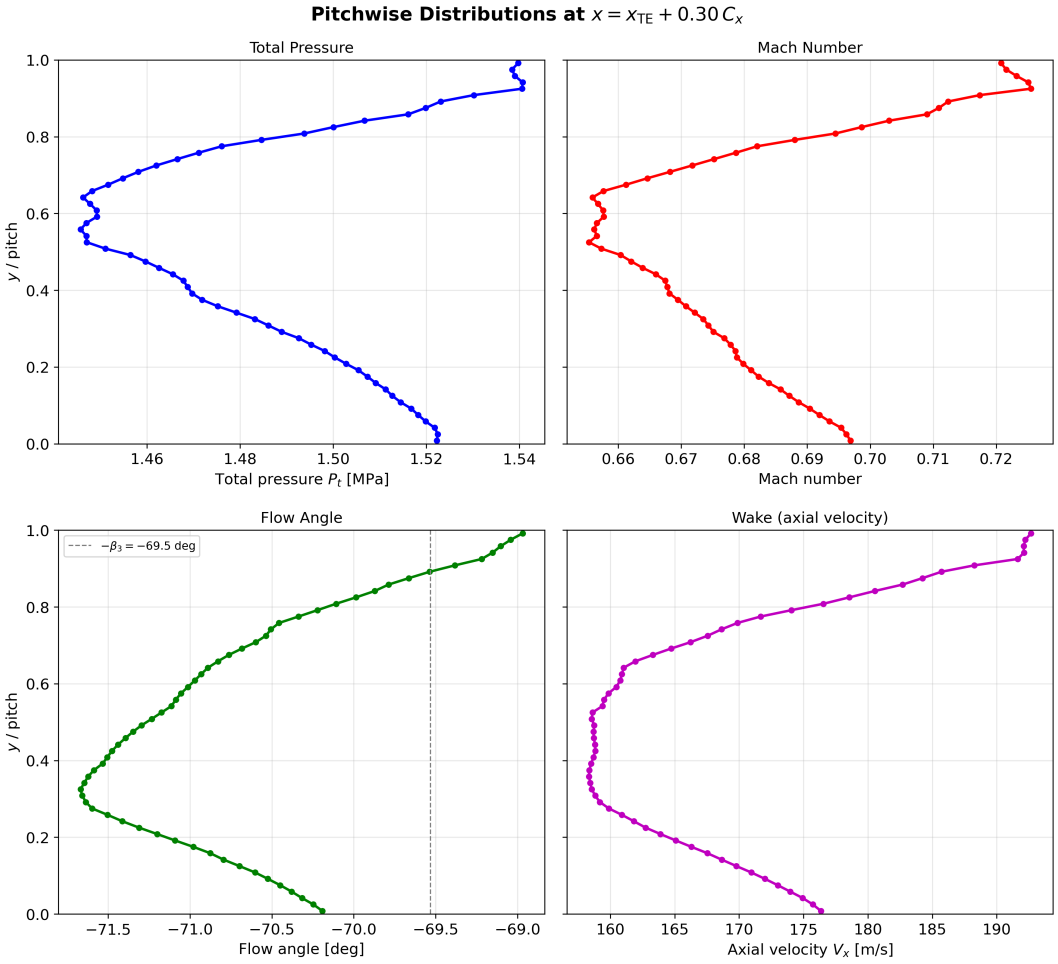


Figure 4.7: Exit-plane pitchwise distributions of total pressure, Mach number, flow angle and axial velocity (wake), shown un-normalised.

4.7 CFD Prediction Result

The CFD confirms the mean-line design: the relative exit Mach number ($M_{w3} \approx 0.67$–0.68) and exit angle ($\approx 70°$ relative) match the design intent, the passage delivers the prescribed expansion to P_3, and the stage total-pressure loss across the rotor is modest ($P_t/P_{t0} = 0.978$, i.e. $\approx 2.2\%$) with a narrow, attached wake. The blade row is therefore aerodynamically sound for the chosen mean-line operating point.

Chapter 5

Conclusion

The first stage of a four-stage high-pressure turbine for an F-class industrial gas turbine (330 MW, 750 kg/s, $TIT = 1400\,^{\circ}\mathrm{C}$) has been designed through the full preliminary-design chain: 1-D mean-line analysis, rotor-blade construction and a cascade CFD evaluation.

The mean-line design, closed by a Newton–Raphson iteration on the exit pressure, delivers the required specific work of 219.27 kJ/kg per stage with a blade speed $U = 349$ m/s at the design coefficients $\psi = 1.8$, $\phi = 0.5$ and $r = 0.35$. The resulting stage has a stator-exit absolute Mach number of 0.70, a rotor turning of 112° and an exit Mach number of 0.30. Of the ten design guidelines, eight are met and two (M_2 and h_3/h_2) are only marginally violated, so the operating point is essentially feasible and could be fully cleared by a small adjustment of the coefficients.

The blade row (42 blades, chord 0.192 m, stagger 39°, $s/c_x = 1.14$) was constructed with PyBlade from the metal angles and shown to reproduce the prescribed curvature. The cascade RANS computation confirms the design: the loading is smooth and front-loaded, the passage delivers the prescribed expansion to $P_3 = 1.107$ MPa, the relative exit Mach number (≈ 0.67) and exit angle ($\approx 70°$) agree with the mean-line prediction, and the rotor total-pressure loss is modest ($\approx 2.2\%$) with a narrow 9.2% wake deficit.

Overall, the design is consistent across the three levels of analysis and forms a sound basis for further refinement (radial equilibrium, cooling, multi-stage matching and a mesh-independence study of the CFD).

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